EXPERIMENT - 10

Q1. Time (seconds) to Days, Hours, Minutes, Seconds

Algorithm

Step 1: Start

Step 2: Input the total time in seconds and store it in variable totalSeconds

Step 3: If totalSeconds is less than 0, display "Invalid input" and go to Step 7

Step 4: Calculate days = totalSeconds / 86400 and remainder rem = totalSeconds % 86400

Step 5: Calculate hours = rem / 3600, update rem = rem % 3600; then minutes = rem / 60 and seconds = rem % 60

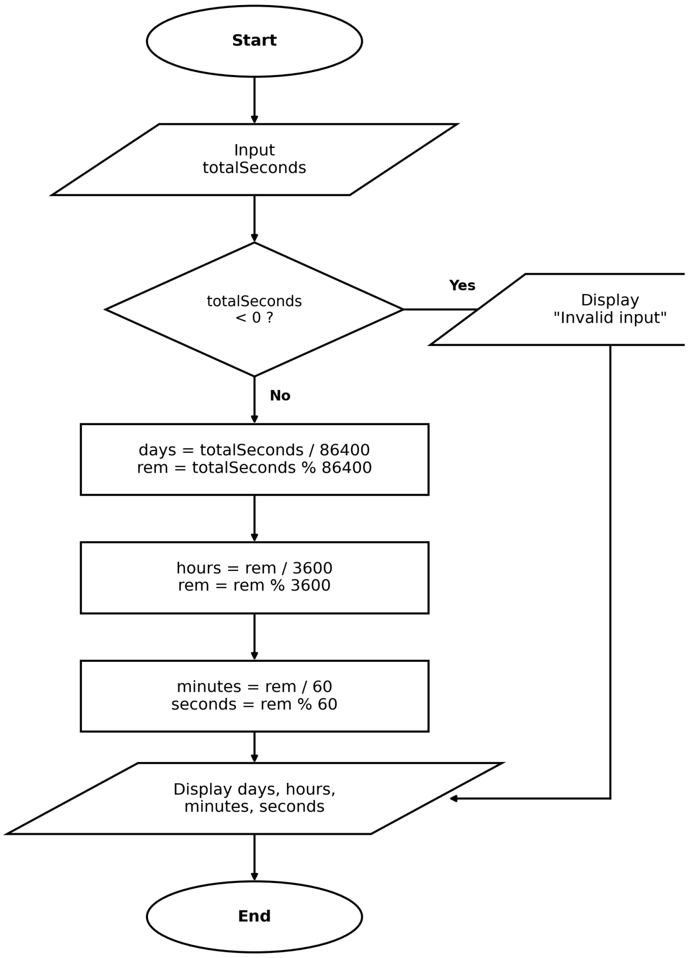
Step 6: Display days, hours, minutes and seconds

Step 7: Stop

Pseudocode

```
BEGIN
    READ totalSeconds
    IF totalSeconds < 0 THEN
        PRINT "Invalid input"
    ELSE
        days     <- totalSeconds DIV 86400
        rem      <- totalSeconds MOD 86400
        hours    <- rem DIV 3600
        rem      <- rem MOD 3600
        minutes  <- rem DIV 60
        seconds  <- rem MOD 60
        PRINT days, hours, minutes, seconds
    ENDIF
END
```

Flowchart



Test Case Table

S.No	Input (seconds)	Expected Output	Actual Output	Result
1	90061	1 Days, 1 Hours, 1 Minutes, 1 Seconds	1 Days, 1 Hours, 1 Minutes, 1 Seconds	**Pass**
2	3661	0 Days, 1 Hours, 1 Minutes, 1 Seconds	0 Days, 1 Hours, 1 Minutes, 1 Seconds	**Pass**
3	0	0 Days, 0 Hours, 0 Minutes, 0 Seconds	0 Days, 0 Hours, 0 Minutes, 0 Seconds	**Pass**
4	59	0 Days, 0 Hours, 0 Minutes, 59 Seconds	0 Days, 0 Hours, 0 Minutes, 59 Seconds	**Pass**
5	-100	Invalid input message	Invalid input! Time in seconds cannot be negative.	**Pass**

C Program

```c
/* HDOC Day-10 : Question 1
   Program to input time in seconds and convert it into
   days, hours, minutes and seconds format.
*/

#include <stdio.h>

int main()
{
    long int totalSeconds, days, hours, minutes, seconds, remaining;

    printf("Enter time in seconds: ");
    scanf("%ld", &totalSeconds);

    if (totalSeconds < 0)
    {
        printf("Invalid input! Time in seconds cannot be negative.\n");
        return 0;
    }

    days = totalSeconds / 86400;
    remaining = totalSeconds % 86400;

    hours = remaining / 3600;
    remaining = remaining % 3600;

    minutes = remaining / 60;
    seconds = remaining % 60;

    printf("Time = %ld Days, %ld Hours, %ld Minutes, %ld Seconds\n",
           days, hours, minutes, seconds);

    return 0;
}
```

Compilation and Execution (Terminal)

```
adilshakil — HDOC_Day10 — -zsh — 120x30

adilshakil@ADILs-MacBook-Air HDOC_Day10 % gcc -o q1_time_converter q1_time_converter.c
adilshakil@ADILs-MacBook-Air HDOC_Day10 % ./q1_time_converter
Enter time in seconds: 90061
Time = 1 Days, 1 Hours, 1 Minutes, 1 Seconds
adilshakil@ADILs-MacBook-Air HDOC_Day10 % ./q1_time_converter
Enter time in seconds: 3661
Time = 0 Days, 1 Hours, 1 Minutes, 1 Seconds
adilshakil@ADILs-MacBook-Air HDOC_Day10 % ./q1_time_converter
Enter time in seconds: 0
Time = 0 Days, 0 Hours, 0 Minutes, 0 Seconds
adilshakil@ADILs-MacBook-Air HDOC_Day10 % ./q1_time_converter
Enter time in seconds: 59
Time = 0 Days, 0 Hours, 0 Minutes, 59 Seconds
adilshakil@ADILs-MacBook-Air HDOC_Day10 % ./q1_time_converter
Enter time in seconds: -100
Invalid input! Time in seconds cannot be negative.
adilshakil@ADILs-MacBook-Air HDOC_Day10 %
```

Q2. Check Leap Year

Algorithm

Step 1: Start

Step 2: Input the year and store it in variable year

Step 3: If year % 4 is not equal to 0, display "Not a Leap Year" and go to Step 6

Step 4: Else if year % 100 is not equal to 0, display "Leap Year" and go to Step 6

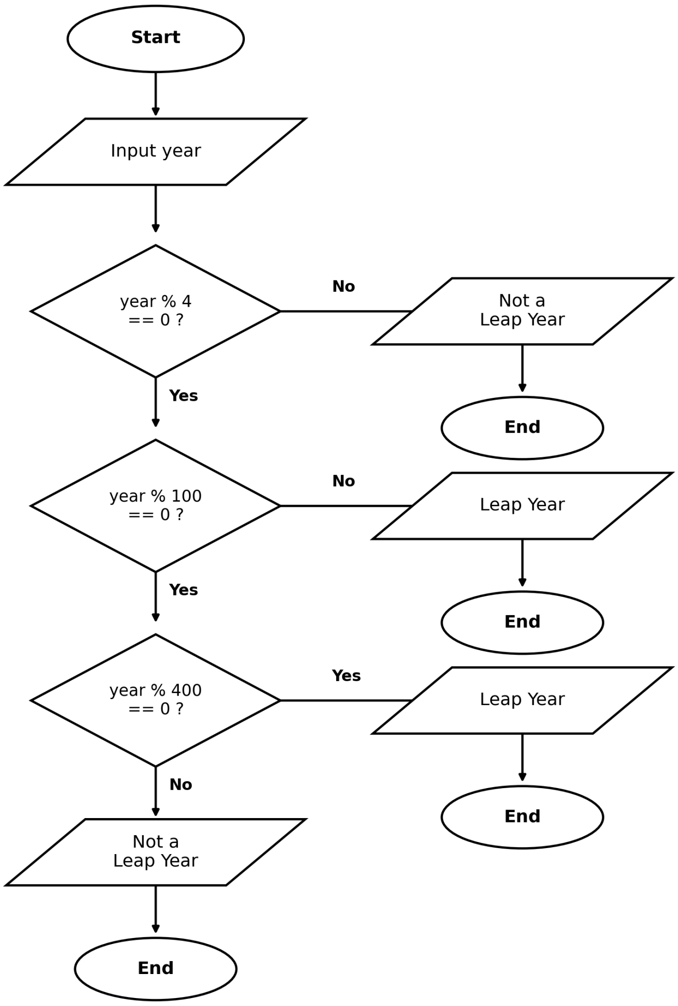
Step 5: Else if year % 400 is equal to 0, display "Leap Year", otherwise display "Not a Leap Year"

Step 6: Stop

Pseudocode

```
BEGIN
    READ year
    IF year MOD 4 = 0 THEN
        IF year MOD 100 = 0 THEN
            IF year MOD 400 = 0 THEN
                PRINT "Leap Year"
            ELSE
                PRINT "Not a Leap Year"
            ENDIF
        ELSE
            PRINT "Leap Year"
        ENDIF
    ELSE
        PRINT "Not a Leap Year"
    ENDIF
END
```

Flowchart



Test Case Table

S.No	Input (year)	Expected Output	Actual Output	Result
1	2000	Leap Year	2000 is a Leap Year	**Pass**
2	1900	Not a Leap Year	1900 is not a Leap Year	**Pass**
3	2024	Leap Year	2024 is a Leap Year	**Pass**
4	2023	Not a Leap Year	2023 is not a Leap Year	**Pass**
5	2400	Leap Year	2400 is a Leap Year	**Pass**

C Program

```c
/* HDOC Day-10 : Question 2
   Program to input a year and check whether it is a
   leap year or not.
   Rule: A year is a leap year if it is divisible by 4,
         but century years (divisible by 100) must also
         be divisible by 400 to be a leap year.
*/

#include <stdio.h>

int main()
{
    int year;

    printf("Enter a year: ");
    scanf("%d", &year);

    if (year % 4 == 0)
    {
        if (year % 100 == 0)
        {
            if (year % 400 == 0)
                printf("%d is a Leap Year\n", year);
            else
                printf("%d is not a Leap Year\n", year);
        }
        else
        {
            printf("%d is a Leap Year\n", year);
        }
    }
    else
    {
        printf("%d is not a Leap Year\n", year);
    }

    return 0;
}
```

Compilation and Execution (Terminal)

adilshakil — HDOC_Day10 — -zsh — 120x30

```
adilshakil@ADILs-MacBook-Air HDOC_Day10 % gcc -o q2_leap_year q2_leap_year.c
adilshakil@ADILs-MacBook-Air HDOC_Day10 % ./q2_leap_year
Enter a year: 2000
2000 is a Leap Year
adilshakil@ADILs-MacBook-Air HDOC_Day10 % ./q2_leap_year
Enter a year: 1900
1900 is not a Leap Year
adilshakil@ADILs-MacBook-Air HDOC_Day10 % ./q2_leap_year
Enter a year: 2024
2024 is a Leap Year
adilshakil@ADILs-MacBook-Air HDOC_Day10 % ./q2_leap_year
Enter a year: 2023
2023 is not a Leap Year
adilshakil@ADILs-MacBook-Air HDOC_Day10 % ./q2_leap_year
Enter a year: 2400
2400 is a Leap Year
adilshakil@ADILs-MacBook-Air HDOC_Day10 %
```